Maximum Subarray Sum

Time limit: 1.00 s
Memory limit: 512 MB

CSES 1643 | Sorting and Searching | https://cses.fi/problemset/task/1643

Given an array of n integers, your task is to find the maximum sum of values in a contiguous, nonempty subarray.

Input

The first input line has an integer n: the size of the array.

The second line has n integers $x_1,x_2,\ldots,x_n$: the array values.

Output

Print one integer: the maximum subarray sum.

Constraints

- $1 \le n \le 2 \cdot 10^5$
- $-10^9 \le x_i \le 10^9$

Example

Input
8 -1 3 -2 5 3 -5 2 2

Output
9